Question 1 – Healthcare Analytics

R Program

```r
R

# Load libraries
library(ggplot2)

# Dataset
health <- data.frame(
  Patient = c("P1","P2","P3","P4","P5","P6","P7","P8"),
  Age = c(24,32,41,48,56,65,38,29),
  BMI = c(19.8,24.5,28.1,30.3,33.2,36.5,26.2,22.1),
  BloodPressure = c(110,118,132,145,152,168,124,116),
  Severity = factor(c("Low","Medium","High","High",
                      "Critical","Critical","Medium","Low"),
                    levels=c("Low","Medium","High","Critical"))
)

# Scatter Plot
ggplot(health,
       aes(x=Age,
           y=BloodPressure,
           color=Severity,
           size=BMI)) +
  geom_point(alpha=0.8) +
  scale_color_manual(values=c(
    "Low"="green",
    "Medium"="yellow",
    "High"="orange",
```

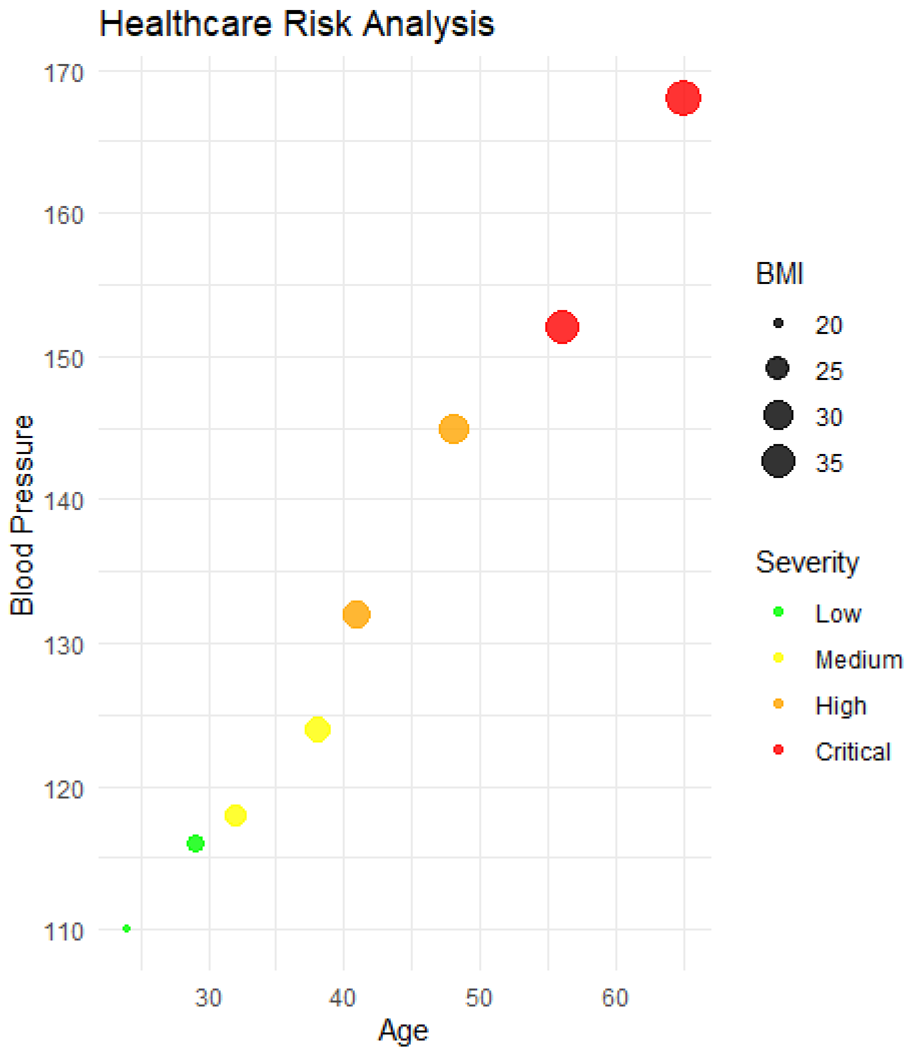
```
  "Critical"="red"
 )) +
 labs(title="Healthcare Risk Analysis",
    x="Age",
    y="Blood Pressure") +
 theme_minimal()
```

Interpretation

- **Age** is placed on the x-axis to observe how blood pressure changes with age.
- **Blood Pressure** on the y-axis directly indicates cardiovascular risk.
- **Disease Severity** is represented using colors:
    - Green → Low
    - Yellow → Medium
    - Orange → High
    - Red → Critical
- **BMI** is shown by point size because larger BMI often indicates higher health risk.

Justification

These mappings allow doctors to identify high-risk patients immediately. Older patients with larger BMI and red-colored points indicate the greatest health concern.



Question 2 – Climate Science

(1) Cartesian Line Chart

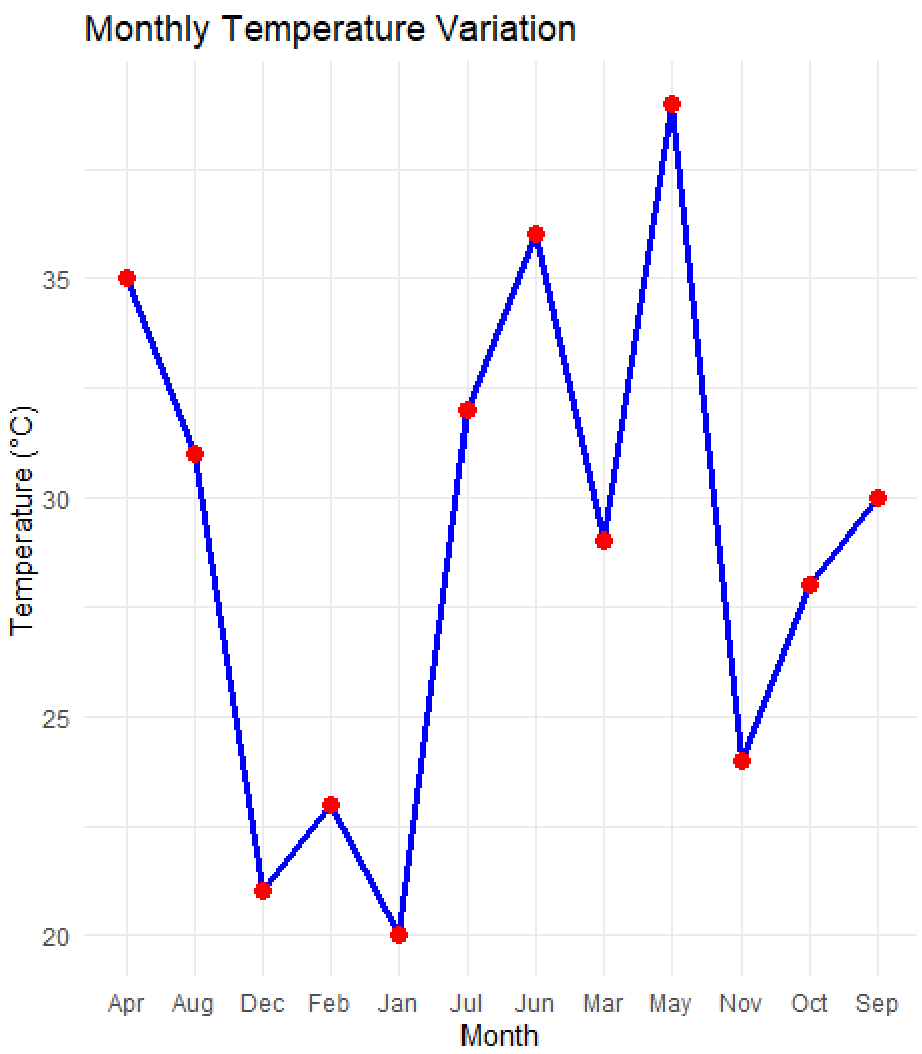
R

```
library(ggplot2)


climate <- data.frame(
 Month=c("Jan","Feb","Mar","Apr","May","Jun",
     "Jul","Aug","Sep","Oct","Nov","Dec"),
```

```r
  SolarRadiation=c(180,200,240,280,320,300,260,250,240,220,190,170),
  Temperature=c(20,23,29,35,39,36,32,31,30,28,24,21),
  Rainfall=c(25,18,12,8,5,65,150,170,120,85,45,30)
)

# Line Chart
ggplot(climate,
       aes(x=Month,
           y=Temperature,
           group=1))+
  geom_line(color="blue",linewidth=1.2)+
  geom_point(size=3,color="red")+
  labs(title="Monthly Temperature Variation",
       x="Month",
       y="Temperature (°C)")+
  theme_minimal()
```



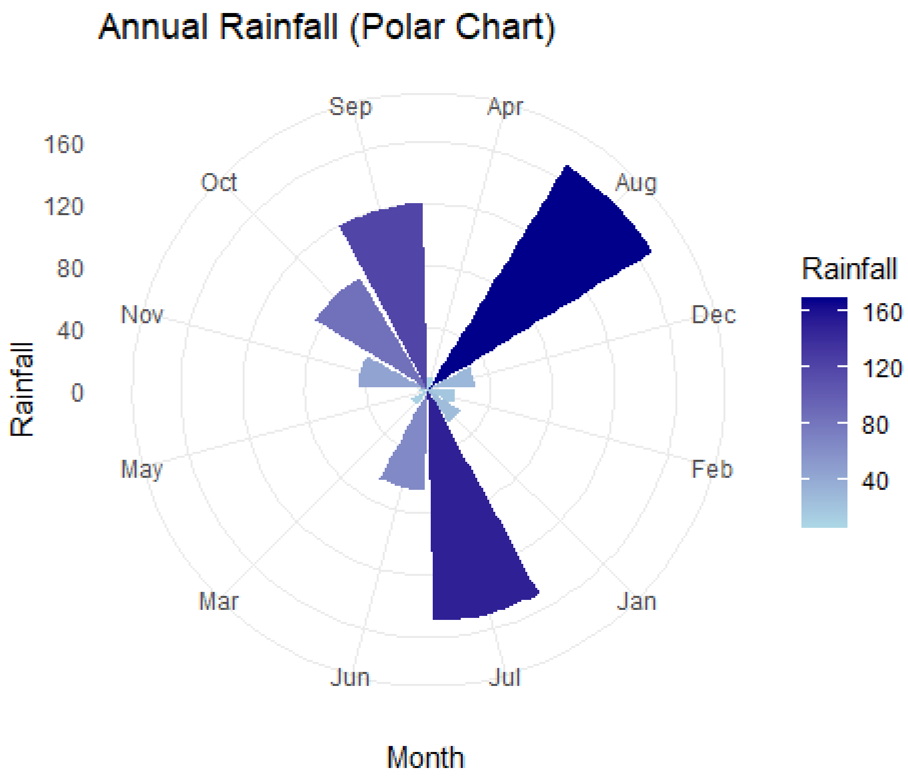
---

(2) Polar Rainfall Chart

R

```
ggplot(climate,
    aes(x=Month,
      y=Rainfall,
      fill=Rainfall))+
 geom_bar(stat="identity")+
 coord_polar()+
 scale_fill_gradient(low="lightblue",
            high="darkblue")+
```

```
labs(title="Annual Rainfall (Polar Chart)")+
theme_minimal()
```



Comparison

Cartesian Line Chart

Advantages

- Shows seasonal trend clearly.
- Easy to compare temperature changes.
- Precise value interpretation.

Limitations

- Does not emphasize yearly cyclic nature.

Polar Chart

Advantages

- Naturally represents annual cycles.
- Highlights seasonal rainfall distribution.

Limitations

- Harder to compare exact rainfall values.
- Less accurate for quantitative comparisons.

Question 3 – Business Analytics & Marketing

R Program

R

```
library(ggplot2)

business <- data.frame(
 Product=c("A","B","C","D","E"),
 Sales=c(210,185,260,145,120),
 Profit=c(18,14,25,12,10),
 MarketShare=c(24,20,28,15,13)
)

business$Highlight <- ifelse(
 business$MarketShare==max(business$MarketShare),
 "Highest","Others"
)

ggplot(business,
    aes(x=Product,
      y=Sales,
      fill=Profit))+
```

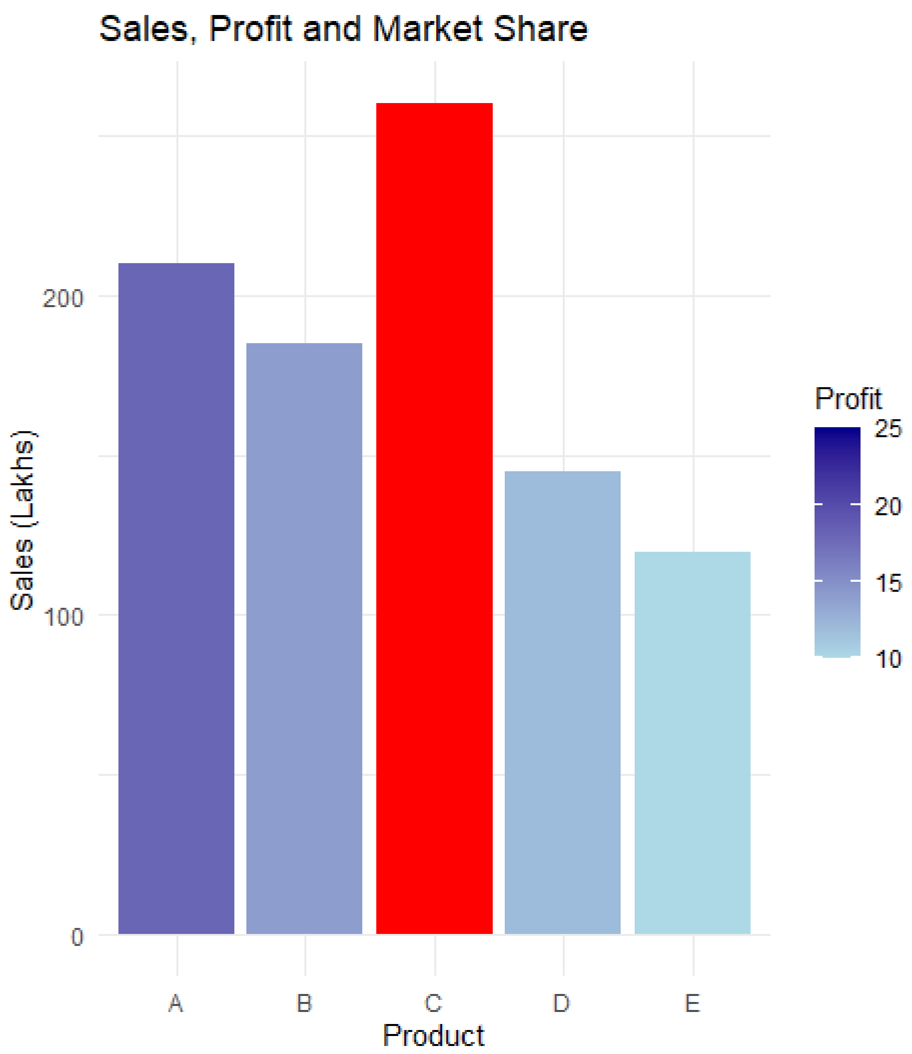
```
geom_bar(stat="identity")+
geom_bar(data=subset(business,Highlight=="Highest"),
     aes(fill=Highlight),
     stat="identity")+
scale_fill_manual(values=c(
  "Highest"="red",
  "10"="#deebf7",
  "12"="#c6dbef",
  "14"="#9ecae1",
  "18"="#6baed6",
  "25"="#2171b5"
),
guide="none")+
labs(title="Sales, Profit and Market Share",
   y="Sales (Lakhs)")+
theme_minimal()
```

Interpretation

- Bar height represents Sales.
- Darker colors indicate higher Profit.
- Product **C** is highlighted in red because it has the highest Market Share.

Justification

Sequential blue shades allow easy comparison of profitability, while the contrasting red immediately identifies the market leader for business decisions.



Question 4 – Transportation Engineering

R Program

R

```
library(ggplot2)


traffic <- data.frame(
 Direction=c("North","North-East","East","South-East",
        "South","South-West","West","North-West"),
 Vehicles=c(980,760,1240,890,1420,1130,860,720)
)
```

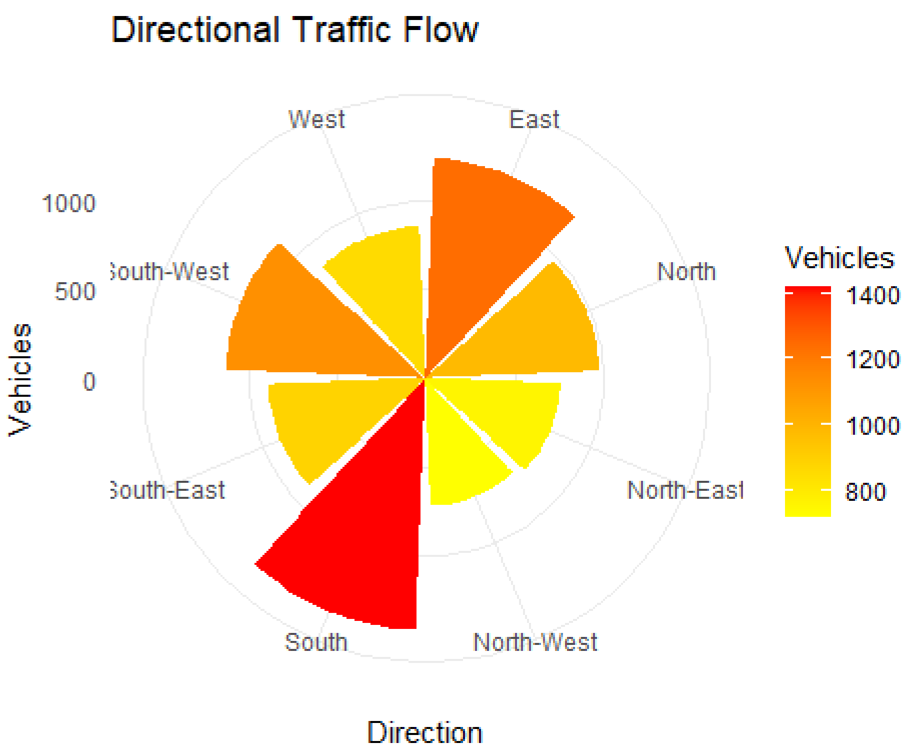
```
ggplot(traffic,
    aes(x=Direction,
      y=Vehicles,
      fill=Vehicles))+
 geom_bar(stat="identity")+
 coord_polar()+
 scale_fill_gradient(low="yellow",
             high="red")+
 labs(title="Directional Traffic Flow")+
 theme_minimal()
```

Interpretation

- Longer bars represent heavier traffic.
- Dark red sectors indicate traffic congestion.
- South has the highest traffic flow.

Why Polar Chart?

- Roads naturally have directional orientation.
- Polar charts preserve directional relationships.
- Easier to identify busiest directions than Cartesian bars.
- Suitable for GIS and transportation analysis.



Question 5 – Environmental Sustainability Dashboard

R Program

R

```
library(ggplot2)

library(gridExtra)


env <- data.frame(

 City=c("A","B","C","D","E","F"),

 AQI=c(48,82,135,165,72,95),

 GreenCover=c(45,32,21,18,38,30),
```

```r
 CO2=c(1.8,3.5,5.6,7.4,2.9,4.2),
 Population=c(9,15,22,28,12,18)
)

env$AQI_Category <- cut(
 env$AQI,
 breaks=c(0,50,100,150,200),
 labels=c("Good","Moderate","Poor","Very Poor")
)

# Scatter Plot
p1 <- ggplot(env,
       aes(x=GreenCover,
         y=CO2,
         color=AQI_Category,
         size=Population))+
 geom_point(alpha=0.8)+
 scale_color_manual(values=c(
  "Good"="green",
  "Moderate"="yellow",
  "Poor"="orange",
  "Very Poor"="red"
 ))+
 labs(title="Green Cover vs CO2")

# AQI Bar Chart
p2 <- ggplot(env,
       aes(x=City,
```

```
          y=AQI,
          fill=AQI_Category))+
  geom_bar(stat="identity")+
  scale_fill_manual(values=c(
   "Good"="green",
   "Moderate"="yellow",
   "Poor"="orange",
   "Very Poor"="red"
  ))+
  labs(title="City AQI")

grid.arrange(p1,p2,ncol=2)
```

Dashboard Justification

Scatter Plot

- **Green Cover (%)** → x-axis
- **CO_2 Emission** → y-axis
- **Population** → point size
- **AQI Category** → color

This shows the relationship between vegetation, pollution, and population.

Bar Chart

Displays AQI comparison across cities.

Color Palette

- 🟢 Good
- 🟡 Moderate
- 🟠 Poor
- 🔴 Very Poor

These colors follow standard environmental pollution indicators, making the dashboard intuitive.

Scales

- Continuous scales for Green Cover, CO_2, Population.
- Categorical color scale for AQI levels.

Overall Design Choice

The dashboard combines multiple environmental indicators into a single view, helping policymakers identify highly polluted cities, understand the relationship between green cover and emissions, and prioritize sustainability initiatives.